

SQL Assessment Answers

Questions 1 and 2 — University Debate Database

Question 1.1 — Create the Tables

```
CREATE TABLE Faculty (  
    FacultyID VARCHAR(4) PRIMARY KEY,  
    FacultyName VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE Venue (  
    VenueID VARCHAR(4) PRIMARY KEY,  
    VenueName VARCHAR(100) NOT NULL,  
    VenueAddress VARCHAR(200) NOT NULL  
);  
  
CREATE TABLE Debate (  
    DebateID VARCHAR(4) PRIMARY KEY,  
    FacultyID_A VARCHAR(4) NOT NULL,  
    FacultyID_B VARCHAR(4) NOT NULL,  
    VenueID VARCHAR(4) NOT NULL,  
    DebateDate DATE NOT NULL,  
    DebateTime TIME NOT NULL,  
    DebateDuration INT NOT NULL,  
  
    CONSTRAINT FK_Debate_Faculty_A  
        FOREIGN KEY (FacultyID_A) REFERENCES Faculty(FacultyID),  
  
    CONSTRAINT FK_Debate_Faculty_B  
        FOREIGN KEY (FacultyID_B) REFERENCES Faculty(FacultyID),  
  
    CONSTRAINT FK_Debate_Venue  
        FOREIGN KEY (VenueID) REFERENCES Venue(VenueID)  
);
```

Question 1.2 — Populate the Tables

```
INSERT INTO Faculty (FacultyID, FacultyName)  
VALUES  
( 'F001', 'Faculty of Science'),  
( 'F002', 'Faculty of Engineering'),  
( 'F003', 'Faculty of Humanities'),  
( 'F004', 'Faculty of Law'),  
( 'F005', 'Faculty of Commerce');  
  
INSERT INTO Venue (VenueID, VenueName, VenueAddress)  
VALUES  
( 'V001', 'Newton Hall', '12 University Road, Cape Town'),  
( 'V002', 'Curie Centre', '44 Innovation Avenue, Gqeberha'),  
( 'V003', 'Plato Auditorium', '88 Knowledge Street, Durban'),  
( 'V004', 'Darwin Hall', '15 Research Lane, Johannesburg'),  
( 'V005', 'Aristotle Theatre', '9 Academic Crescent, Tshwane');  
  
INSERT INTO Debate  
(DebateID, FacultyID_A, FacultyID_B, VenueID,  
    DebateDate, DebateTime, DebateDuration)  
VALUES  
( 'D001', 'F001', 'F002', 'V001', '2026-08-10', '10:00:00', 60),  
( 'D002', 'F003', 'F005', 'V004', '2026-08-11', '14:00:00', 90),  
( 'D003', 'F002', 'F003', 'V005', '2026-08-12', '18:00:00', 90),  
( 'D004', 'F001', 'F005', 'V003', '2026-08-13', '16:00:00', 120),  
( 'D005', 'F002', 'F005', 'V001', '2026-08-14', '17:00:00', 120);
```

Question 1.3 — Add SeatsAvailable

```
ALTER TABLE Debate
```

```
ADD SeatsAvailable INT;
```

Question 1.4 — Update D003

```
UPDATE Debate
SET SeatsAvailable = 400
WHERE DebateID = 'D003';
```

Question 2.1 — Venues With No Debates

```
SELECT v.VenueName
FROM Venue v
LEFT JOIN Debate d
    ON v.VenueID = d.VenueID
WHERE d.DebateID IS NULL;
```

Question 2.2 — Total Duration at Each Venue

```
SELECT
    v.VenueName,
    SUM(d.DebateDuration) AS Total_Duration
FROM Venue v
LEFT JOIN Debate d
    ON v.VenueID = d.VenueID
GROUP BY v.VenueID, v.VenueName
ORDER BY v.VenueName ASC;
```

Question 2.3 — Longest Debate at V001

```
SELECT
    v.VenueName,
    d.DebateDate,
    d.DebateDuration
FROM Debate d
JOIN Venue v
    ON d.VenueID = v.VenueID
WHERE d.VenueID = 'V001'
    AND d.DebateDuration = (
        SELECT MAX(DebateDuration)
        FROM Debate
        WHERE VenueID = 'V001'
    );
```

Expected Results

Question 2.1

VENUE_NAME
Curie Centre

Question 2.2

VENUE_NAME	TOTAL_DURATION
Aristotle Theatre	90
Curie Centre	NULL
Darwin Hall	90
Newton Hall	180
Plato Auditorium	120

Question 2.3

VENUE_NAME	DEBATE_DATE	DEBATE_DURATION
Newton Hall	2026-08-14	120